

# Representations of Lie Groups

## A Mathematical Dissection of Quantum Spin

### Part I: Mathematical Foundations

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- 3 Smooth Manifolds
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## Definition (topological space)

A **topological space** is a pair  $(X, \mathcal{T})$ , where  $X$  is a set and  $\mathcal{T} \subseteq \mathcal{P}(X)$  satisfies:

- ① (Trivial bounds)  $\emptyset \in \mathcal{T}$  and  $X \in \mathcal{T}$ .
- ② (Arbitrary unions)  $\forall \{U_\alpha\}_{\alpha \in \mathcal{I}} \subseteq \mathcal{T} : \bigcup_{\alpha \in \mathcal{I}} U_\alpha \in \mathcal{T}$ .
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**Key constructions.** Product topology, subspace topology, metric topologies (e.g. the Euclidean topology on  $\mathbb{R}^n$ ).

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**Why Hausdorff + second-countable?** These are technical conditions that prevent pathological spaces. They guarantee that partitions of unity exist.



## Definition (smooth manifold)

A **smooth manifold** is a triple  $(M, \mathcal{T}, \mathcal{A})$  where:

- ①  $(M, \mathcal{T})$  is a topological manifold.
- ②  $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{I}}$  is a **smooth atlas**: each  $(U_\alpha, \varphi_\alpha)$  is a chart, the charts cover  $M$ , and transition maps  $\varphi_\beta \circ \varphi_\alpha^{-1}$  are  $C^\infty$ .
- ③  $\mathcal{A}$  is **maximal** with respect to smooth compatibility.

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- 3  $\mathcal{A}$  is **maximal** with respect to smooth compatibility.

**Key idea.** Charts are local coordinate systems. Smoothness means that when you switch between coordinate systems, the change-of-coordinates map is infinitely differentiable.

# Smooth Maps and Diffeomorphisms

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A map  $F : M \rightarrow N$  between smooth manifolds is **smooth** if its coordinate representation  $\psi \circ F \circ \varphi^{-1}$  is  $C^\infty$  for all compatible charts.

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**Example.**  $\mathbb{R}^n$  is diffeomorphic to the open unit ball  $B_1(0)$ .  $S^n$  is *not* diffeomorphic to  $\mathbb{R}^n$  (compact vs. non-compact).

# Tangent Spaces via Derivations

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How do we define tangent vectors to a smooth manifold without embedding it in  $\mathbb{R}^N$ ? Answer: **derivations** — linear maps that satisfy the Leibniz rule.

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Let  $M$  be a smooth manifold,  $p \in M$ . The **stalk**  $C_p^\infty M$  is the set of germs of smooth functions at  $p$ . The **tangent space** at  $p$  is

$$T_p M \stackrel{\text{df}}{=} \{v : C_p^\infty M \rightarrow \mathbb{R} \mid v \text{ is a derivation at } p\}.$$

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**Why derivations?** For  $\mathbb{R}^n$ , a tangent vector  $v$  acts on functions by the directional derivative:  $v(f) = \sum_i v^i \frac{\partial f}{\partial x^i} \Big|_p$ . This algebraic characterisation generalises to any manifold.



# The Differential

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Let  $F : M \rightarrow N$  be smooth,  $p \in M$ . The **differential of  $F$  at  $p$**  is the linear map  $\mathbf{d}F_p : T_p M \rightarrow T_{F(p)} N$  defined by

$$\forall v \in T_p M, \forall [f]_{F(p)} \in C_{F(p)}^\infty N : \quad [(\mathbf{d}F_p(v))]([f]_{F(p)}) \stackrel{\text{df}}{=} v([f \circ F]_p).$$

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## Chain rule

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**Interpretation.** The differential is the “best linear approximation” to a smooth map, generalising the Jacobian matrix from multivariable calculus.

# Groups, Fields, and Vector Spaces

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**Why these matter.** Lie groups combine the smooth structure of manifolds with the algebraic structure of groups. Lie algebras are vector spaces with an extra bracket operation.

# Algebras and Lie Algebras

## Definition ( $\mathbb{F}$ -algebra)

A vector space  $\mathcal{A}$  over  $\mathbb{F}$  together with a bilinear multiplication  $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ . (Examples:  $\mathbb{R}^{n,n}$  with matrix multiplication,  $C^\infty(M)$  with pointwise multiplication.)



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## Definition (Lie algebra)

An  $\mathbb{F}$ -vector space  $\mathfrak{g}$  with a bilinear bracket  $[\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$  satisfying:

- ① (Skew-symmetry)  $[Y, X] = -[X, Y]$
- ② (Jacobi identity)  $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$

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**Key example.**  $(\mathbb{R}^3, \times)$  — the cross product makes  $\mathbb{R}^3$  a Lie algebra!  
 $[v, w] = v \times w$  satisfies skew-symmetry and Jacobi.

# Lie Groups

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A **Lie group** is a smooth manifold  $(G, \mathcal{T}, \mathcal{A})$  that is also a group  $(G, \{\mu, \iota, e\})$  such that:

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## Examples.

- $(\mathbb{R}^n, +)$  — the translation group.
- $\mathrm{GL}(n, \mathbb{R})$  — invertible  $n \times n$  real matrices.
- $\mathrm{SO}(n)$  — rotations in  $\mathbb{R}^n$ .
- $\mathrm{SU}(n)$  — unitary transformations with determinant 1.

# Left Translation and Left-Invariant Vector Fields

## Definition (left translation)

For  $g \in G$ ,  $L_g : G \rightarrow G$  is defined by  $L_g(h) \stackrel{\text{df}}{=} gh$ .  $L_g$  is a diffeomorphism.

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**Geometric meaning.** A left-invariant vector field is completely determined by its value at the identity  $e$ . Translating the vector at  $e$  by  $L_g$  gives the vector at any point  $g$ .



# The Lie Algebra of a Lie Group

## Theorem (tangent space at identity is a Lie algebra)

Let  $G$  be a Lie group. Define  $\mathfrak{g} = T_e G$  (tangent space at identity). For each  $v \in \mathfrak{g}$ , there exists a **unique** left-invariant vector field  $X^v$  with  $X^v|_e = v$ . Define the bracket on  $\mathfrak{g}$  by

$$[v, w]_{\mathfrak{g}} \stackrel{\text{df}}{=} [X^v, X^w]_{\circ}|_e.$$

Then  $(\mathfrak{g}, [\bullet, \bullet]_{\mathfrak{g}})$  is a Lie  $\mathbb{R}$ -algebra with  $\dim(\mathfrak{g}) = \dim(G)$ .

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**Remark.** We denote the Lie algebra of a Lie group  $G$  by  $\text{Lie}(G)$ .

# The Closed Subgroup Theorem (É. Cartan)

## Theorem (Closed Subgroup Theorem)

Let  $G$  be a Lie group and  $H \subseteq G$  a closed subgroup. Then  $H$  admits a unique smooth structure making it a Lie group such that the inclusion  $H \hookrightarrow G$  is a smooth embedding.

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**Example.**  $\mathrm{SO}(n)$ ,  $\mathrm{SU}(n)$ ,  $\mathrm{SL}(n, \mathbb{R})$ , and  $\mathrm{Sp}(2n)$  are all closed subgroups of  $\mathrm{GL}(N, \mathbb{F})$  and therefore Lie groups.

# The General Linear Group $\text{GL}(n, \mathbb{F})$

Example (the general linear group)

$$\text{GL}(n, \mathbb{F}) = \{\mathbf{A} \in \mathbb{F}^{n,n} : \det(\mathbf{A}) \neq 0\}.$$

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- Its Lie algebra is  $\mathfrak{gl}(n, \mathbb{F}) = \mathbb{F}^{n,n}$  (all  $n \times n$  matrices) with the commutator bracket:

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- Most matrix Lie groups arise as closed subgroups of  $\mathrm{GL}(n, \mathbb{F})$ . Their Lie algebras are subspaces of  $\mathfrak{gl}(n, \mathbb{F})$  closed under the commutator.

# The Special Orthogonal Group $\text{SO}(3)$

Example (the special orthogonal group — rigid rotations)

$$\text{SO}(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1 \};$$

$$\mathfrak{so}(3) = T_{\mathbf{I}_3} \text{SO}(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^\top = -\mathbf{R} \}.$$

$\dim(\mathfrak{so}(3)) = \dim(\text{SO}(3)) = 3$ , with basis  $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$ :

$$\mathbf{L}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad \mathbf{L}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

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**Remark.**  $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$  — the same structure as the cross product on  $\mathbb{R}^3$ .

# The Special Unitary Group $SU(2)$

## Example (the special unitary group)

$$SU(2) = \left\{ U \in \mathbb{C}^{2,2} : U^\dagger U = I_2, \det(U) = 1 \right\};$$

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$$T_1 = \frac{1}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad T_2 = \frac{1}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad T_3 = \frac{1}{2} \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}.$$

# The Special Unitary Group $SU(2)$

## Example (the special unitary group)

$$SU(2) = \left\{ U \in \mathbb{C}^{2,2} : U^\dagger U = I_2, \det(U) = 1 \right\};$$

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**Remark.**  $[T_i, T_j] = \varepsilon_{ijk} T_k$  — **identical** to  $\mathfrak{so}(3)$ ! This is the first hint of the isomorphism  $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ .

# The Double Cover $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$

## Construction

For  $U \in \mathrm{SU}(2)$ , define  $\Phi(U) : \mathfrak{su}(2) \rightarrow \mathfrak{su}(2)$  by

$$[\Phi(U)](X) \stackrel{\mathrm{df}}{=} UXU^\dagger.$$

Using the basis  $\beta = (T_1, T_2, T_3)$ , identify  $\mathfrak{su}(2) \cong \mathbb{R}^3$ . Then the matrix  $[\Phi(U)]_\beta^\beta$  lies in  $\mathrm{SO}(3)$ .

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## Theorem (double cover $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ )

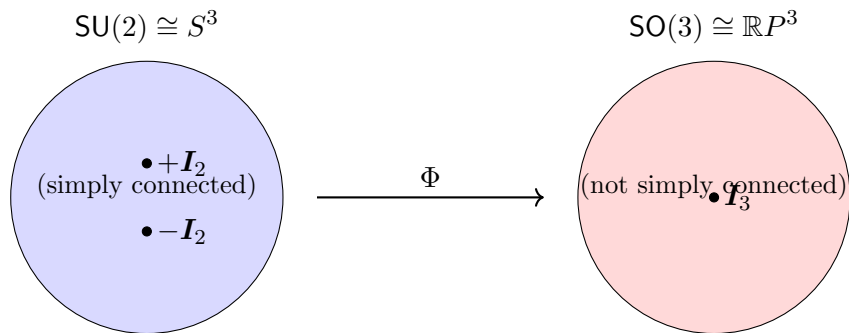
$\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  is:

- ① A smooth group homomorphism:  $\Phi(UV) = \Phi(U) \circ \Phi(V)$ .
- ② Surjective.
- ③ Has kernel  $\ker(\Phi) = \{\pm I_2\}$  (two elements).

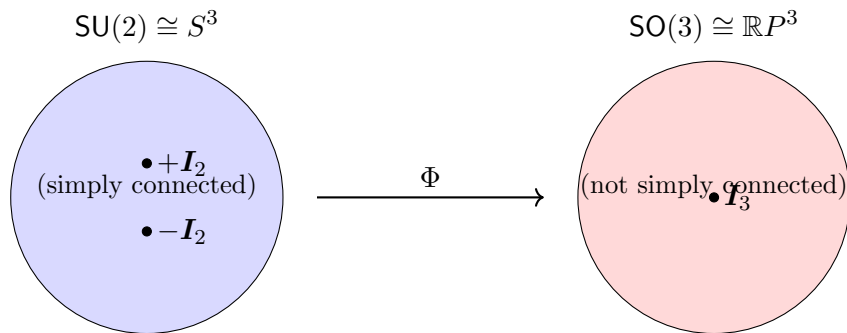
Thus  $\mathrm{SU}(2)$  is a **double cover** of  $\mathrm{SO}(3)$ .



# The Double Cover: Geometric Picture



# The Double Cover: Geometric Picture



**Why two-to-one?** Both  $U$  and  $-U$  give the same rotation. A rotation by angle  $\theta$  about axis  $\hat{n}$  lifts to  $\pm \cos(\theta/2)I_2 \mp i \sin(\theta/2)(\hat{n} \cdot \sigma)$ . Increasing  $\theta$  by  $2\pi$  flips the sign!

# Lie Algebra Isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$

## Theorem (Lie algebra isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ )

Define  $\varphi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$  by  $\varphi(\mathbf{L}_i) \stackrel{\text{df}}{=} \mathbf{T}_i$  (and extend  $\mathbb{R}$ -linearly). Then  $\varphi$  is a Lie algebra isomorphism.

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## Proof (essence)

- $\varphi$  maps a basis to a basis  $\implies$  linear isomorphism.
- Both satisfy  $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$  and  $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk} \mathbf{T}_k$ .
- Therefore
$$\varphi([\mathbf{L}_i, \mathbf{L}_j]) = \varphi(\varepsilon_{ijk} \mathbf{L}_k) = \varepsilon_{ijk} \mathbf{T}_k = [\mathbf{T}_i, \mathbf{T}_j] = [\varphi(\mathbf{L}_i), \varphi(\mathbf{L}_j)].$$

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**Significance.** Although  $\text{SU}(2)$  and  $\text{SO}(3)$  are different as groups (double cover vs. quotient), their Lie algebras are **identical**. Infinitesimal rotations are the same in both pictures.

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- 6  $SO(3)$  (rotations) and  $SU(2)$  (unitary  $2 \times 2$  matrices) are the central examples.
- 7  $SU(2)$  is a **double cover** of  $SO(3)$ . Their Lie algebras are **isomorphic**:  $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ .

# What Comes Next (Part II)

## Representation theory (next meeting)

- Representations of Lie algebras and Lie groups.
- Complexification  $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$  and the Cartan–Weyl basis  $(\mathbf{H}, \mathbf{E}, \mathbf{F})$ .
- Classification of irreducible representations of  $\mathfrak{su}(2)$ : for each  $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ , a unique irrep of dimension  $2j + 1$ .
- Integration: every  $\mathfrak{su}(2)$ -representation exponentiates to an  $SU(2)$ -representation.
- Projective representations and the descent to  $SO(3)$ : integer  $j \rightarrow$  ordinary representations, half-integer  $j \rightarrow$  spinor (projective) representations.
- Physical spin operators and the  $4\pi$  periodicity of spinors.

- Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction* (2nd ed.). Springer.
- Lee, J. M. (2012). *Introduction to Smooth Manifolds* (2nd ed.). Springer.
- Steinacker, H. (2024). *Lie Groups and Lie Algebras for Physicists*. Lecture notes, University of Vienna.
- Full paper with proofs: [github.com/ganddoffcha/quantum-spin](https://github.com/ganddoffcha/quantum-spin)

Thank you!

Questions?